Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack engineer with 2+ years building **Python/FastAPI backends**, **LLM-powered applications**, and **ETL pipelines** deployed on Kubernetes. Experienced shipping production AI workflows with GPT-4, Claude, and async distributed systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **FastAPI microservices** serving 500+ users with async background workers, reducing API latency from **1.2s to 350ms** through query optimization and efficient PostgreSQL access patterns.
- Developed **ETL pipelines** processing 5K+ monthly jobs across SQL databases and cloud storage with validation checks, incremental loads, and error recovery, achieving 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **LLM-powered agentic workflows** integrating GPT-4o, Claude 3.7, and vision models with switchable architecture, processing 10K+ daily requests with token tracking and cost-aware model selection.
- Developed **full-stack features using Flask and React** with concurrent batch processing, implementing async file ingestion and REST APIs while maintaining stable performance under production workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **data pipelines processing 100K+ records** using Python and PyTorch, implementing feature extraction, batch processing, and experiment tracking with Weights & Biases for reproducible ML workflows.
- Designed **transformer-based models** with multi-head attention and spatial-temporal encoding, improving planning accuracy by 18% through systematic experimentation and performance benchmarking.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **ETL workflows with AWS Glue and S3**, implementing data transformation pipelines with incremental loads, improving pipeline efficiency by **68%** and reducing storage costs by 15%.
- Built **CI/CD pipelines with Jenkins** for containerized deployments, maintaining 20+ automated test suites with PyTest and achieving 68% improvement in build reliability across environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **event-driven microservices** with 8 services for authentication, resource allocation, and notifications, handling 2M+ records with WebSocket-based real-time synchronization.
- Deployed **Docker containers on Azure** with Jenkins CI/CD, implementing Kubernetes orchestration basics, staged deployments, and production monitoring, cutting release cycles by **50%**.

Skills

Languages: **Python**, TypeScript, JavaScript, SQL, Java, Bash
Backend: FastAPI, Flask, SQLAlchemy, async services, REST APIs, event-driven architecture
Frontend: React, Next.js, Vue basics, TypeScript interfaces, responsive web applications
Data & AI: PostgreSQL, Snowflake basics, ETL pipelines, LLM orchestration (GPT, Claude), PyTorch
Infrastructure: Docker, Kubernetes, Helm basics, Terraform basics, AWS, Azure, CI/CD, Git

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an **LLM-powered document platform** with semantic search and conversational Q&A across 50K+ documents, achieving 95% query accuracy through vector embeddings and retrieval-augmented generation.

PiXMod Live | *Next.js, MongoDB, Transformers, Stripe*
- Built an **AI-powered SaaS platform** using CLIP and Stable Diffusion for image manipulation, implementing semantic search across 100K+ embeddings with Stripe billing and async processing workflows.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Built a **containerized data processing API** deployed on AWS ECS/Fargate with CodePipeline automation, implementing validation gates and production-grade reliability across deployment environments.